Delivery Performance SLA Analysis

Olist E-Commerce Dataset

Business Intelligence & Data Analysis Report

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Business Question:
Delivery Performance SLA Analysis

Tools Used:
Python (Pandas, NumPy, Matplotlib, Seaborn)

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1. Introduction

Delivery performance is one of the most critical factors affecting customer satisfaction in e-commerce. Meeting the promised delivery date enhances customer trust, while delays may lead to negative reviews and reduced customer loyalty.

This report analyzes the Delivery Performance Service Level Agreement (SLA) using the Olist e-commerce dataset. The objective is to evaluate the company's delivery performance, identify the main factors contributing to delivery delays, and assess how these delays influence customer satisfaction. The analysis also provides business insights and recommendations to support operational improvements and enhance logistics efficiency.

2. Business Question

How well does Olist meet its Delivery Performance Service Level Agreement (SLA), and what are the main factors affecting delivery delays and customer satisfaction?

3. Data Preparation

3.1 Data Cleaning

The analysis was performed using the **orders** dataset from the Olist e-commerce database.

Before conducting the analysis, all date-related columns were converted to the datetime format to enable accurate time calculations. Missing values were identified and evaluated to ensure data quality.

The original dataset contained **99,441** orders with different order statuses. Since the objective of this analysis is to evaluate delivery performance, only completed orders with the status **"delivered"** were retained. After filtering, the final dataset contained **96,478 delivered orders**.

3.2 Feature Engineering

Several performance indicators were created to evaluate the delivery process:

- Delivery Time
- Delay
- Approval Time
- Shipping Time
- Carrier Time

These indicators were used throughout the analysis to evaluate delivery efficiency and identify the major sources of delay.

3.3 Delay Segmentation

Delayed orders were classified into four categories:

- On Time / Early
- Average Week (1–7 days)
- Average Month (8–30 days)
- More Than Month (>30 days)

This classification helps measure the severity of delivery delays and supports business decision-making.

4. Analysis and Results

4.1 On-Time Delivery Performance

To evaluate Olist's compliance with the Delivery Service Level Agreement (SLA), the percentage of orders delivered on time was calculated by comparing the actual delivery date with the estimated delivery date.

Results

- **Total Delivered Orders:** 96,478
- **On-Time Delivery Rate: 93.22%**
- **Delayed Delivery Rate: 6.77%**

Figure 1. Distribution of On-Time and Delayed Deliveries.

These results indicate that Olist maintains a high level of delivery performance and successfully meets its delivery commitments for the vast majority of customer orders. However, the delayed deliveries represent an opportunity for further improvement, particularly by investigating the causes of delays and optimizing logistics operations.

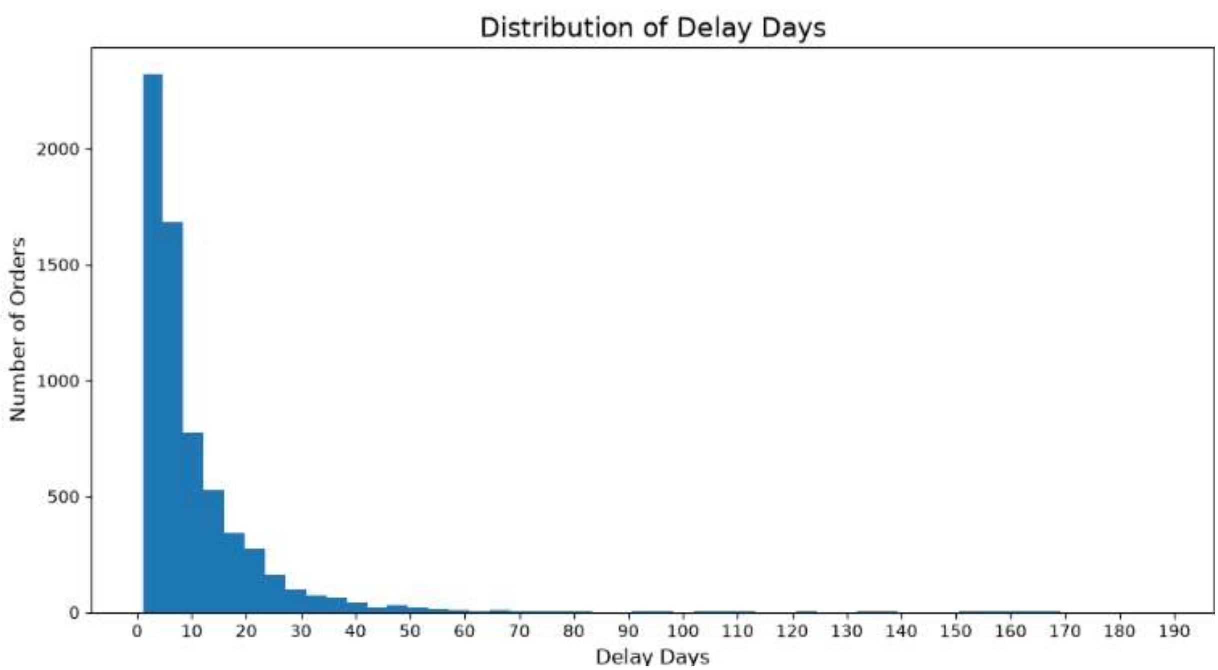
4.2 Distribution of Delay Days

To understand the severity of delayed deliveries, the number of delay days was analyzed for all late orders.

The histogram shows that most delayed orders experienced only a few days of delay, with the highest concentration occurring within the first 10 days after the estimated delivery date. As the number of delay days increases, the number of affected orders decreases significantly. Only a small number of orders experienced very long delays exceeding one month.

Business Insight:
The majority of delivery issues are minor and can likely be reduced through improvements in logistics coordination and shipment scheduling. Extreme delays are rare but should be investigated individually, as they may indicate exceptional operational problems.

Figure 2. Distribution of Delay Days



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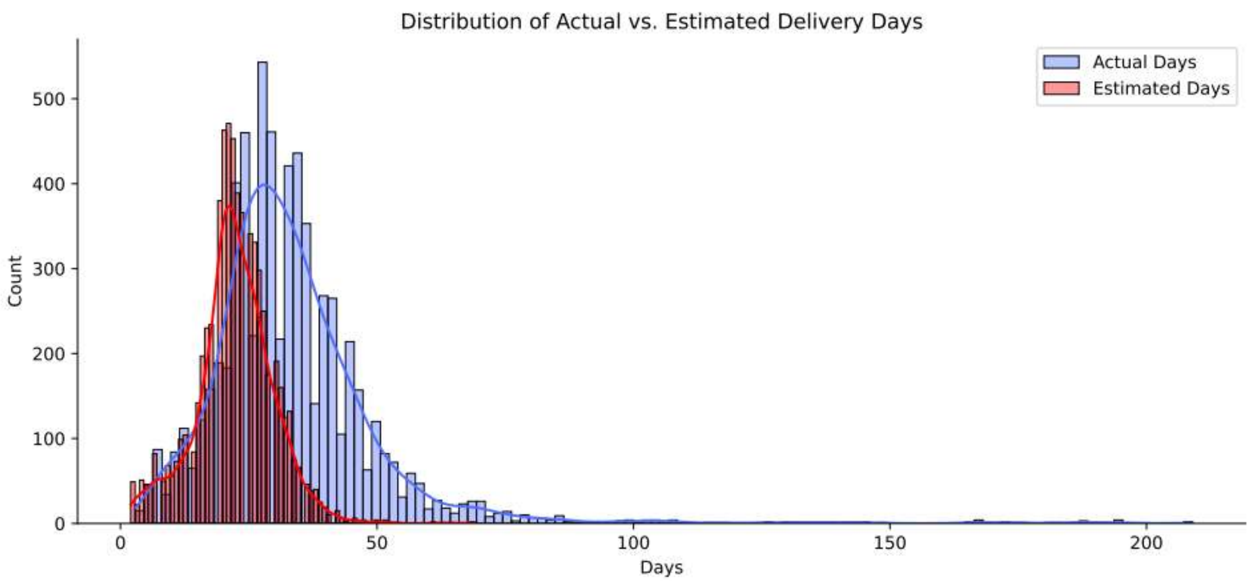
4.3 Actual vs. Estimated Delivery Time

The distribution of actual delivery days was compared with the estimated delivery days to evaluate the accuracy of Olist's delivery estimates.

The comparison shows that the actual delivery time generally follows the expected delivery pattern, although it exhibits a wider distribution and a longer right tail. This indicates that while most deliveries are completed close to the estimated timeframe, a small proportion of orders require considerably more time than expected.

Business Insight:
The estimated delivery dates are generally reliable for most customers. However, the long-tail distribution suggests that certain orders are affected by exceptional delays, highlighting opportunities to improve logistics performance and reduce delivery variability.

Figure 3. Distribution of Actual vs. Estimated Delivery Days.



4.4 Delay Categories Distribution

To better understand the severity of delayed deliveries, late orders were grouped into three categories based on the number of delay days:

- **Average Week:** 1–7 days
- **Average Month:** 8–30 days
- **More Than Month:** More than 30 days

Results

The analysis shows that:

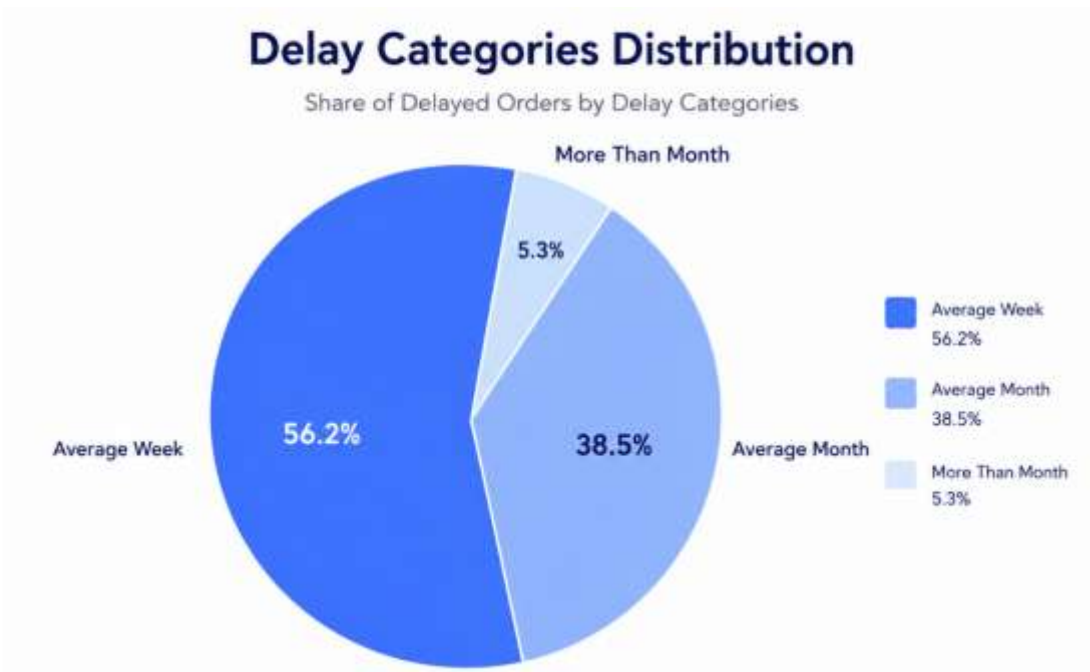
- **56.2%** of delayed orders were delivered within **one week** after the estimated delivery date.
- **38.5%** experienced delays ranging from **8 to 30 days**.
- Only **5.3%** of delayed orders were delayed for **more than one month**.

The shortest recorded delay was **1 day**, while the longest reached **188 days**.

Business Insight

Most delivery delays are relatively short, indicating that operational issues are generally resolved within a few days. However, although delays exceeding one month represent a small percentage of cases, they may have a significant negative impact on customer satisfaction and should be investigated individually to identify their root causes.

Figure 4. Distribution of Delay Categories.



4.5 Average Time Across Delivery Stages

To identify which stage contributes the most to delivery delays, the average duration of each stage in the delivery process was calculated for delayed orders.

Results

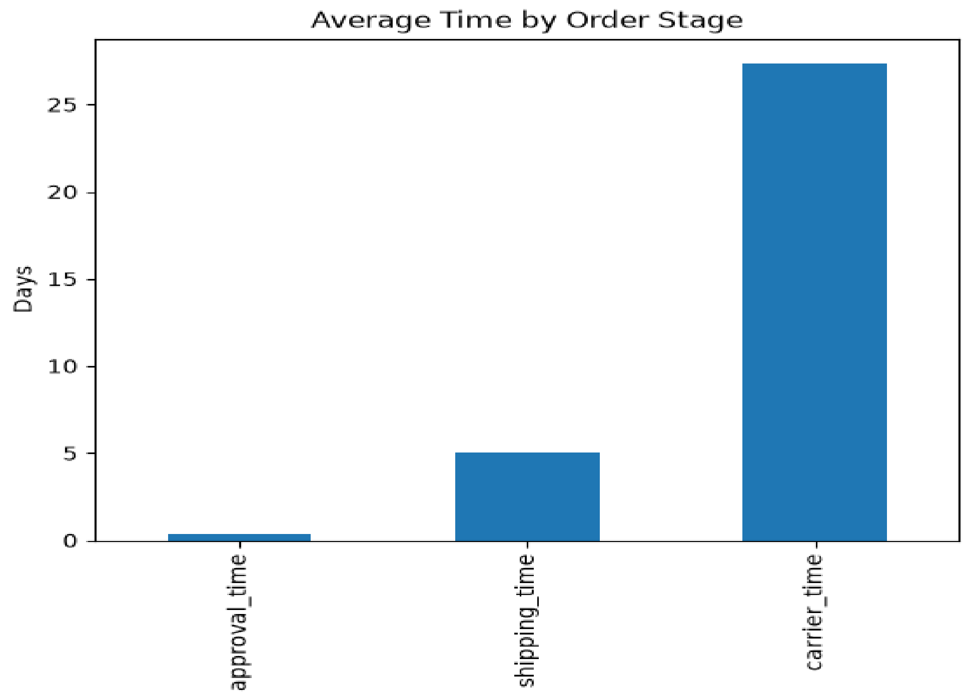
Delivery Stage	Average Time (Days)
Order Approval	0.33
Shipping to Carrier	5.05
Carrier to Customer	27.39

The results indicate that the **carrier-to-customer stage** is by far the longest phase in the delivery process, with an average duration of **27.39 days**. In comparison, order approval takes less than one day on average, while shipping the order to the carrier requires approximately **5.05 days**.

Business Insight

These findings suggest that the main contributor to delayed deliveries is the transportation stage between the carrier and the customer rather than the order approval or shipment preparation processes. Therefore, improving carrier performance, optimizing delivery routes, and strengthening logistics partnerships could significantly reduce delivery delays and enhance overall service quality.

Figure 5. Average Time Across Delivery Stages.



4.6 Seller Performance by State

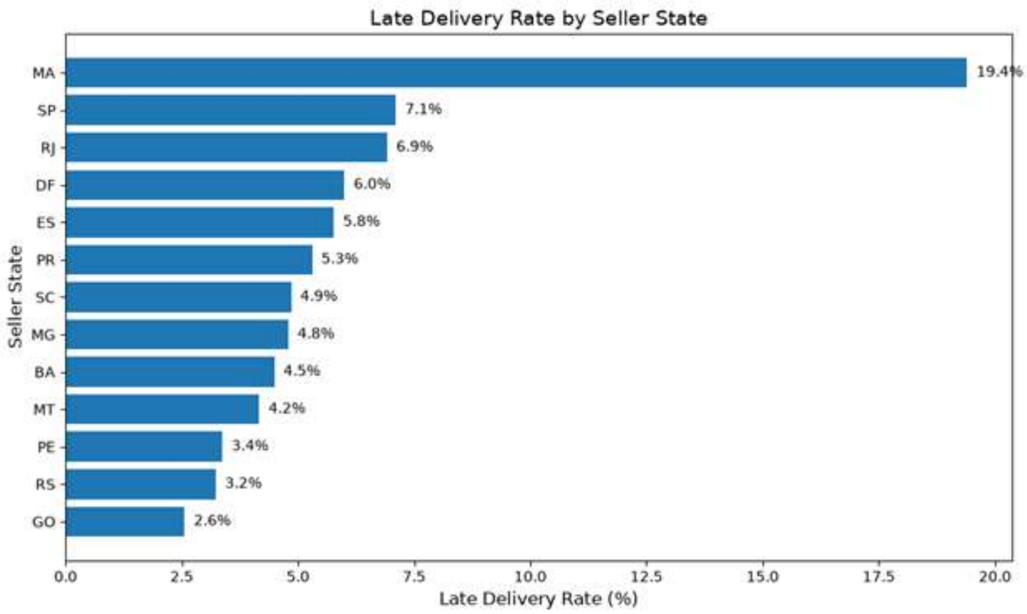
To investigate whether delivery performance varies by seller location, the orders dataset was merged with the order items, sellers, and customers datasets. The late delivery rate was then calculated for each seller state.

Results

The analysis revealed noticeable differences in delivery performance across seller states.

Seller State	Total Orders	Late Delivery Rate (%)
MA	402	19.40
SP	78,604	7.11
RJ	4,685	6.92
DF	883	6.00
ES	364	5.77
PR	8,487	5.30
SC	4,000	4.85
MG	8,603	4.79
BA	624	4.49
MT	144	4.17
PE	445	3.37
RS	2,169	3.23
GO	508	2.56

Figure 6.1. Late Delivery Rate by Seller State.



Business Insight

The results show that seller performance differs across states. **MA** recorded the highest late delivery rate (**19.40%**), although it handled a relatively small number of orders. In contrast, **SP**, which processed the largest order volume, maintained a considerably lower late delivery rate (**7.11%**). This suggests that operational efficiency and logistics management play a more significant role than order volume alone.

These findings can help Olist identify regions that require additional operational support and improve delivery performance by working more closely with sellers and logistics partners in high-delay states.

Top Sellers with Highest Late Delivery Rate

To evaluate the impact of individual sellers on delivery performance, the late delivery rate was calculated for sellers with more than 100 completed orders.

Results

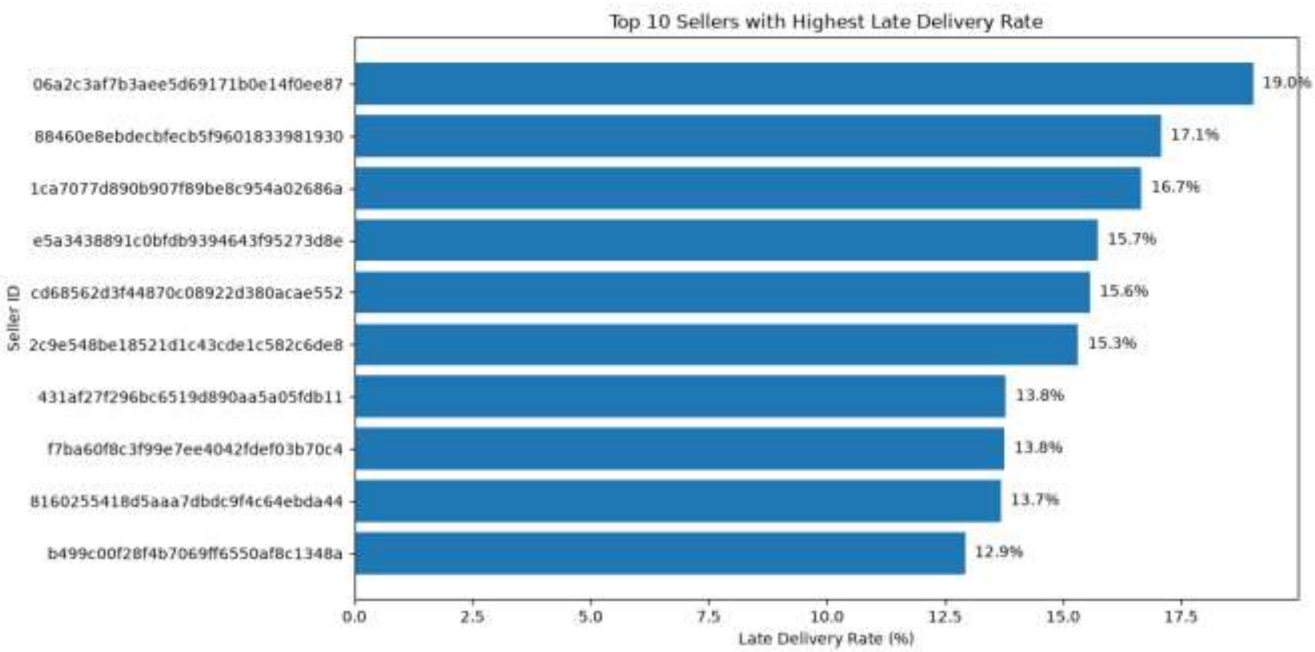
The analysis revealed significant differences in delivery performance among sellers. While most sellers maintained relatively low late delivery rates, a few sellers showed considerably higher percentages of delayed deliveries.

The seller with ID **06a2c3af7b3aee5d69171b0e14f0ee87** recorded the highest late delivery rate (**19.02%**), followed by seller **8160255418d5aaa7dbdc9f4c64ebda44** (**13.68%**) and seller **7d13fca15225358621be4086e1eb0964** (**11.47%**).

Business Insight

The variation in delivery performance indicates that seller-related operations can significantly influence delivery outcomes. Identifying sellers with consistently high late delivery rates enables Olist to provide operational support, monitor performance, and implement corrective actions to improve customer satisfaction and overall logistics efficiency.

Figure 6.2. Top 10 Sellers with Highest Late Delivery Rate.



4.7 Relationship Between Seller Order Volume and Late Delivery Rate

To examine whether seller workload affects delivery performance, the relationship between the total number of completed orders and the late delivery rate was analyzed using a scatter plot.

Results

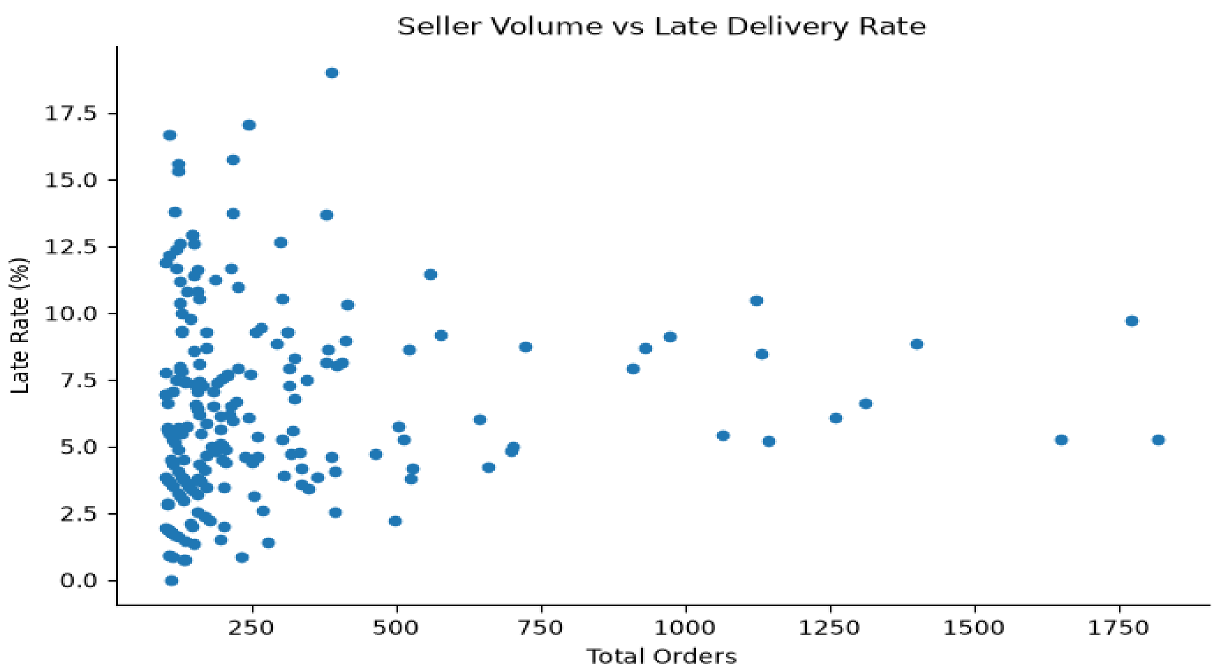
The scatter plot shows that sellers with both low and high order volumes exhibit varying late delivery rates. Most sellers are concentrated between **100 and 400 completed orders**, with late delivery rates generally ranging from **2% to 10%**. A few sellers with relatively low order volumes experienced exceptionally high late delivery rates, exceeding **15%**.

Large-volume sellers generally maintained moderate late delivery rates, suggesting that handling more orders does not necessarily result in poorer delivery performance.

Business Insight

The results indicate that there is **no strong relationship** between the number of orders handled by a seller and the percentage of delayed deliveries. This suggests that delivery performance is influenced more by operational efficiency and logistics management than by seller order volume alone. Therefore, Olist should focus on monitoring seller performance and improving logistics processes rather than assuming that high-volume sellers are more likely to experience delivery delays.

Figure 7. Relationship Between Seller Order Volume and Late Delivery Rate.



4.8 Relationship Between Freight Cost and Delivery Delay

To examine whether higher shipping costs lead to better delivery performance, the relationship between freight value and delivery delay was analyzed using a scatter plot.

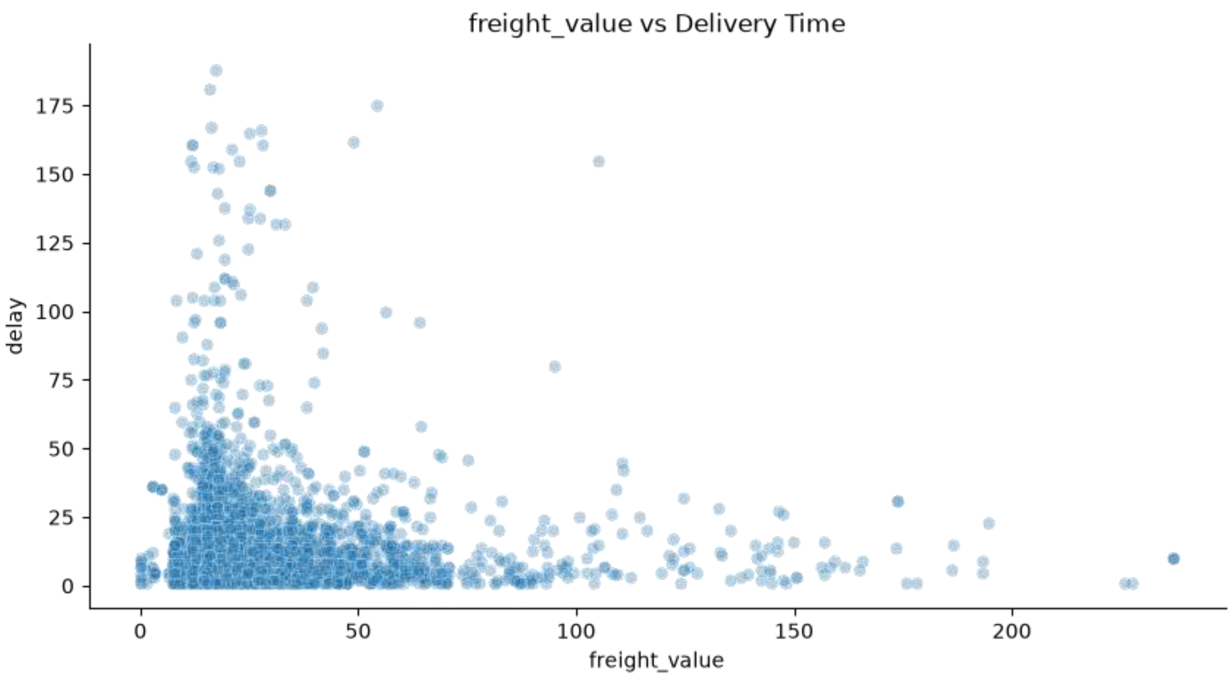
Results

The scatter plot shows a wide dispersion of delay values across different freight costs, with no clear linear relationship between the two variables. Most orders are concentrated at relatively low freight values, while both short and long delivery delays occur across different shipping cost levels.

Business Insight

The analysis suggests that paying a higher freight cost does not necessarily guarantee faster delivery or lower delivery delays. This indicates that delivery performance is influenced by multiple operational factors, such as seller processing time, carrier efficiency, warehouse operations, and logistics management, rather than shipping cost alone.

Figure 8. Relationship Between Freight Value and Delivery Delay.



4.9 Impact of Delivery Performance on Customer Satisfaction

To evaluate the impact of delivery performance on customer satisfaction, customer review scores were compared between on-time and delayed orders.

Results

Customer review scores showed a significant difference based on delivery performance.

- **On-Time Orders:** Average Review Score = **4.29**
- **Delayed Orders:** Average Review Score = **2.27**

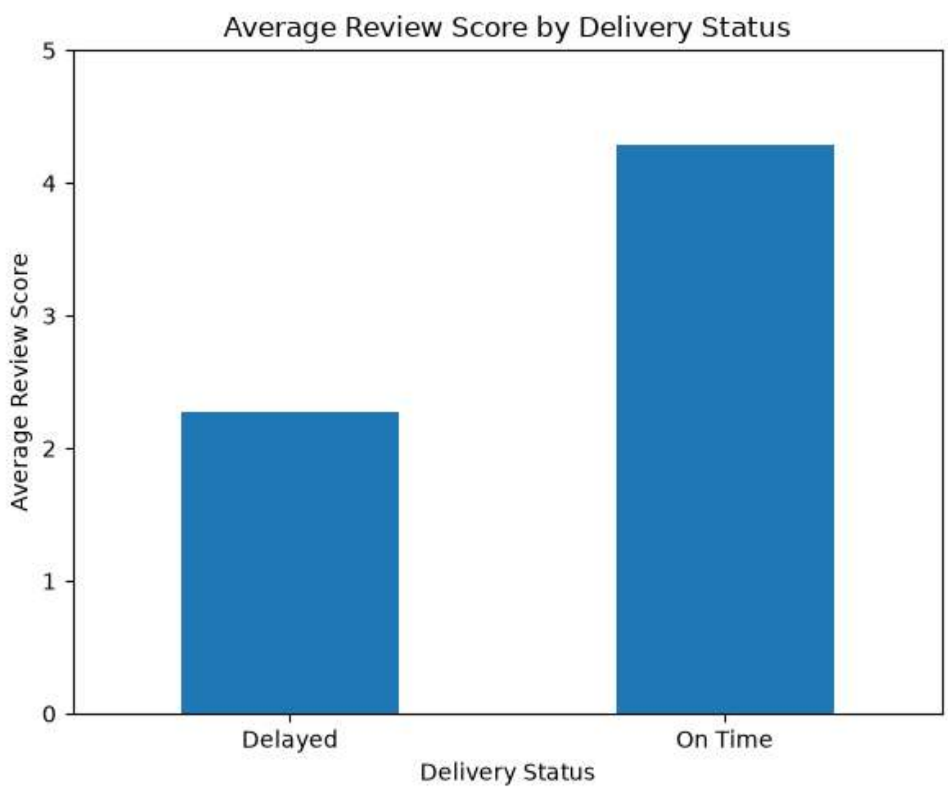
The analysis revealed a rating drop of approximately **2.02 stars** for delayed orders compared to on-time orders.

Business Insight

The results demonstrate that delivery performance has a strong influence on customer satisfaction. Customers whose orders were delivered on time gave significantly higher ratings than those who experienced delivery delays.

A decrease of more than **2 stars** indicates that delayed deliveries negatively affect the customer experience and may reduce customer loyalty, increase negative reviews, and damage the company's reputation. Therefore, improving delivery reliability should remain a strategic priority for Olist to enhance customer satisfaction and encourage repeat purchases.

Figure 9. Average Review Score by Delivery Status.



5. Key Findings

The main findings of the Delivery Performance SLA analysis are summarized below:

- Olist achieved an **on-time delivery rate of 93.22%**, while only **6.77%** of delivered orders were delayed.
- More than half of the delayed orders were delivered within **one week** after the estimated delivery date, while only a small percentage experienced delays longer than one month.
- The **carrier-to-customer** stage was identified as the longest phase of the delivery process, indicating that transportation is the primary contributor to delivery delays.
- Delivery performance varied across seller states, suggesting regional differences in logistics efficiency.
- Shipping time and carrier time showed a **moderate positive correlation** with late delivery rates, indicating that longer operational times are associated with higher delivery delays.
- Delivery distance showed **almost no relationship** with delivery delays, suggesting that operational efficiency has a greater impact than geographical distance.
- Freight value showed **no significant relationship** with delivery delays, indicating that higher shipping costs do not necessarily result in faster deliveries.
- Customer satisfaction was strongly affected by delivery performance. Delayed orders received an average review score of **2.27**, compared with **4.29** for on-time deliveries, representing a decrease of approximately **2.02 stars**.

6. Recommendations

Based on the analysis results, the following recommendations are proposed to improve Olist's Delivery Performance SLA:

- Improve carrier performance by monitoring transportation efficiency and reducing carrier delivery time.
- Monitor sellers with consistently high late delivery rates and provide operational support where necessary.
- Optimize warehouse and shipping preparation processes to reduce shipping time before carrier pickup.
- Continuously monitor Delivery SLA performance using business dashboards and key performance indicators (KPIs).
- Investigate orders with exceptionally long delivery delays to identify and eliminate root causes.
- Improve communication with customers by providing timely delivery updates and notifications when delays occur.
- Focus operational improvements on logistics efficiency rather than increasing freight costs, as higher shipping costs were not associated with better delivery performance.
- Continue improving delivery reliability to maintain high customer satisfaction and encourage repeat purchases.

7. Conclusion

This report evaluated Olist's Delivery Performance Service Level Agreement (SLA) using operational and customer review data. The analysis showed that Olist maintains a high on-time delivery rate, demonstrating strong overall delivery performance. However, opportunities for improvement remain, particularly in carrier operations, shipping efficiency, and seller performance. Furthermore, delivery delays were found to have a significant negative impact on customer satisfaction. Implementing the recommendations presented in this report can help reduce delivery delays, improve logistics performance, enhance customer satisfaction, and strengthen long-term customer loyalty.

8. Appendix

Performance Indicators Used

1. Delivery Time

Measures the total number of days between the order purchase date and the actual customer delivery date.

Formula:
Delivery Time = Order Delivered Customer Date − Order Purchase Timestamp

2. Delivery Delay

Measures the difference between the actual delivery date and the estimated delivery date.

Formula:
Delay = Order Delivered Customer Date − Estimated Delivery Date

- Delay ≤ 0 → On-Time / Early
- Delay > 0 → Delayed

3. Approval Time

Measures the time required for order approval.

Formula:
Approval Time = Order Approved At − Order Purchase Timestamp

4. Shipping Time

Measures the time between order approval and shipment to the carrier.

Formula:
Shipping Time = Order Delivered Carrier Date − Order Approved At

5. Carrier Time

Measures the transportation time from the carrier to the customer.

Formula:
Carrier Time = Order Delivered Customer Date − Order Delivered Carrier Date

6. On-Time Delivery Rate

Formula:

On-Time Delivery Rate = (Number of On-Time Orders / Total Delivered Orders) × 100

7. Late Delivery Rate

Formula:

Late Delivery Rate = (Number of Delayed Orders / Total Delivered Orders) × 100

8. Seller Late Delivery Rate

Formula:

Seller Late Rate = (Seller Delayed Orders / Seller Total Orders) × 100

9. Pearson Correlation Coefficient

Measures the strength and direction of the linear relationship between two variables. The coefficient ranges from **-1** to **+1**.

- +1: Perfect positive correlation
- 0: No linear correlation

- -1: Perfect negative correlation

10. Delivery Distance (Haversine Formula)

The distance between seller and customer locations was calculated using the Haversine formula based on latitude and longitude coordinates.

11. Average Review Score

Represents the average customer satisfaction rating for delivered orders, calculated from the review scores associated with each order.